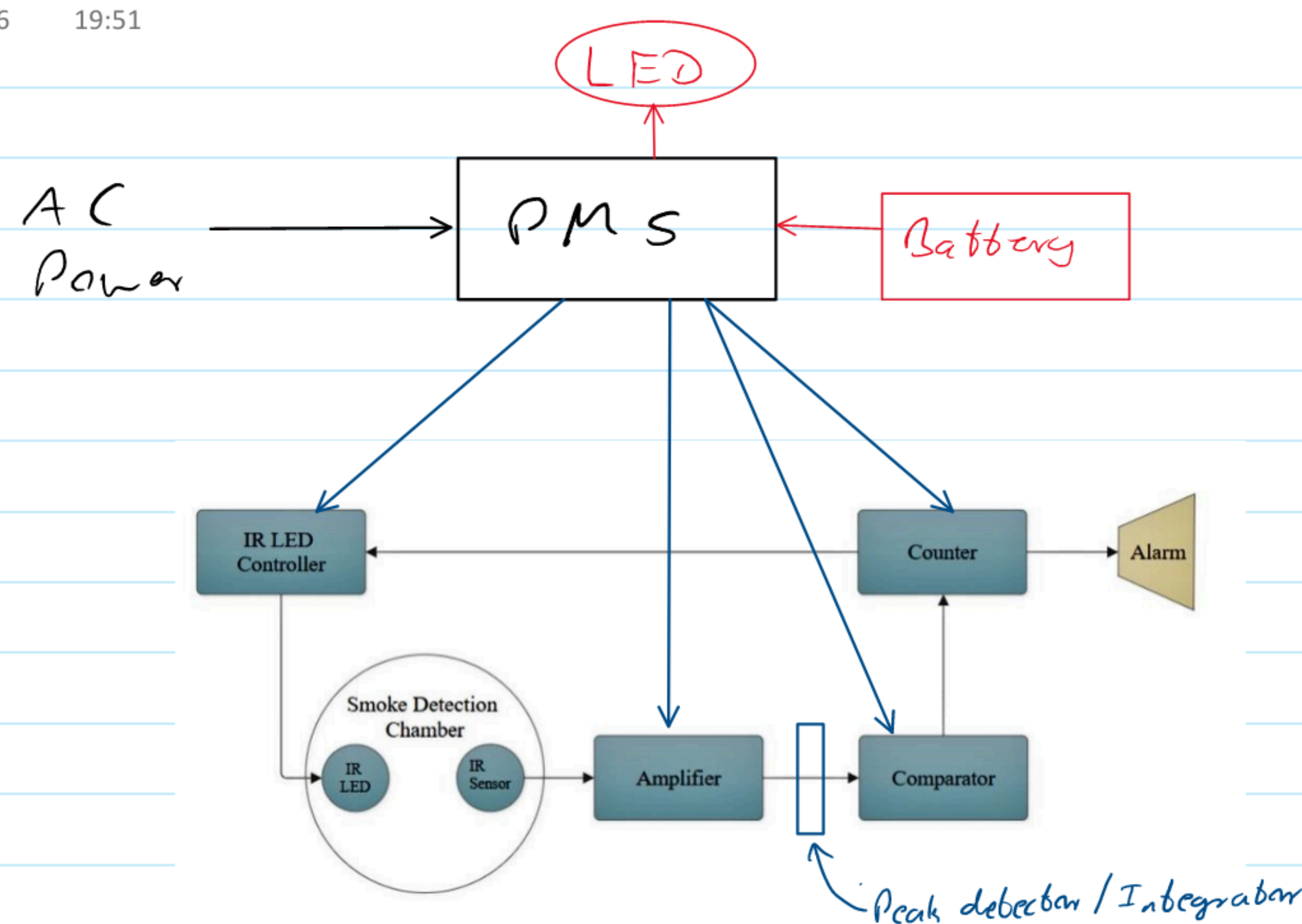




Product Specs

- 1. Detects smoke and triggers alarm
- 2. The alarm should stop if the smoke goes away
- 3. Must handle hysteresis
- 4. Must indicate proper operation

Possible options

- 1. LED connected to PMS, lighted in concerns
- 2. LED parallel to IR LED, not blinking means issues

Need to decide. May depend on the power supply

- AC only
- AC + Battery
- Battery only
 - Rechargeable
 - Non rechargeable

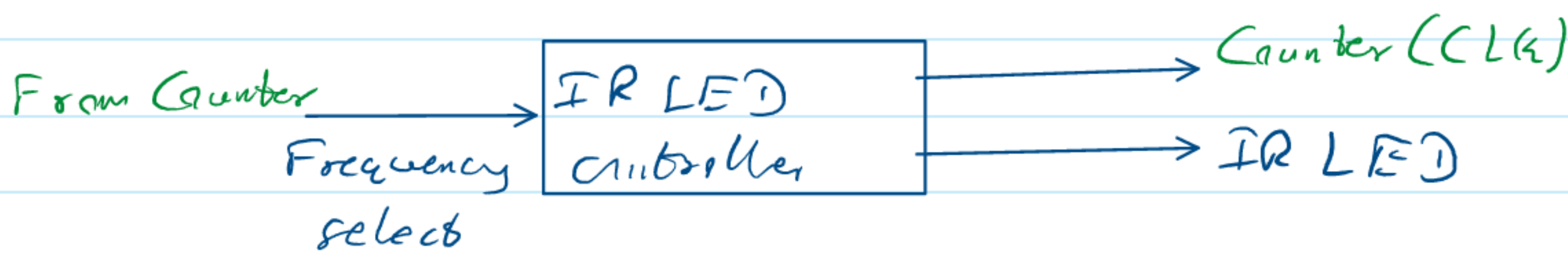
Which one?
(May depend on the supervisor)

Where is the oscillator?

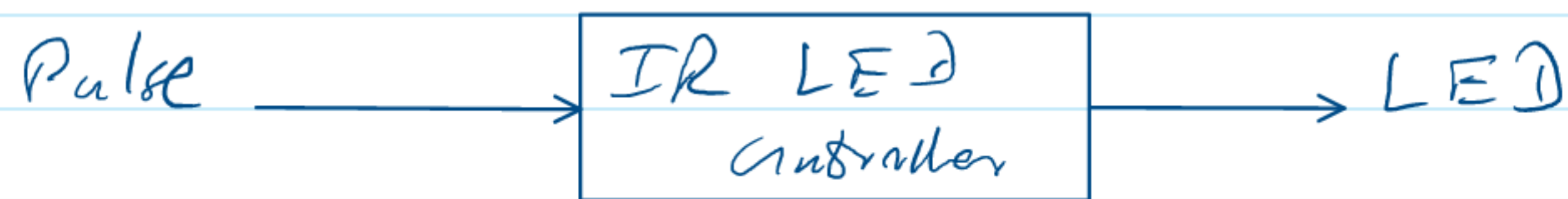
Option 1 - In the IR LED Controller

Option 2 - Oscillator in the counter

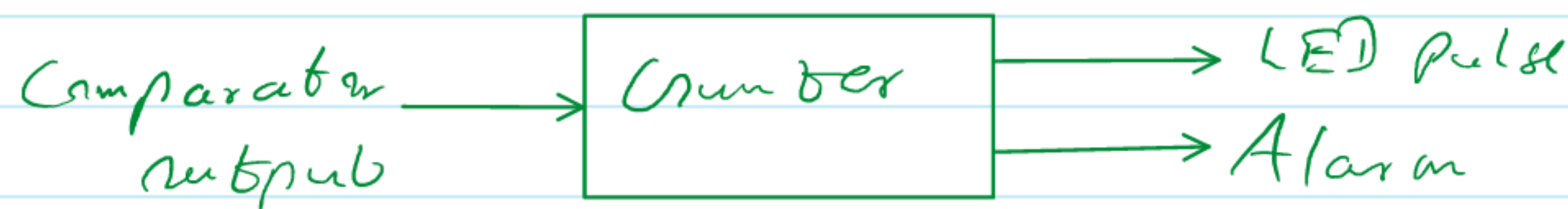
1. IR LED controller



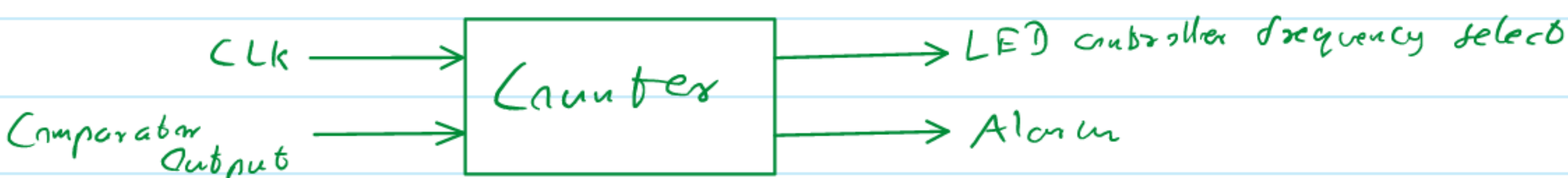
- Frequency select determines the frequency
Normal operation - Low (eg:- once in 30s)
Once a high pulse is detected - high (eg:- 1Hz)
- Oscillator duty cycle should be as low as possible to save power.



Or merge this to the counter itself



2. Counter



Initially
Counter = 00
LED amb. freq. select = Low
Alarm = Low

At each clock cycle;
If Counter != 11
If comparator = High
Increment counter
If Counter != 00
If comparator = Low
Decrement counter

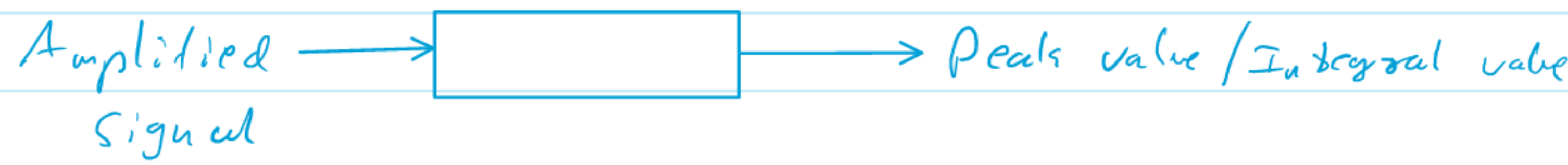
Asynchronous;
If Counter != 00
LED amb. freq. select = High
If Counter = 11
Alarm = High
If Counter = 00
Alarm = Low
Else
Alarm unchanged

3. Amplifier



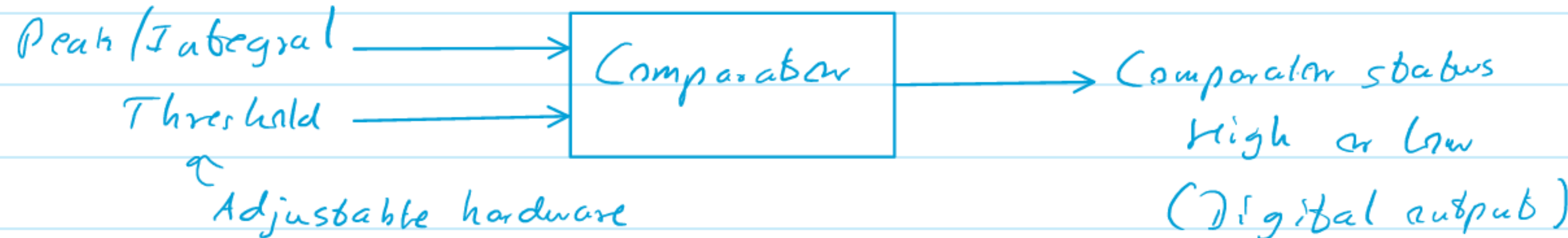
Amplifies whatever coming from the photodiode

4. Peak detector or Integrator



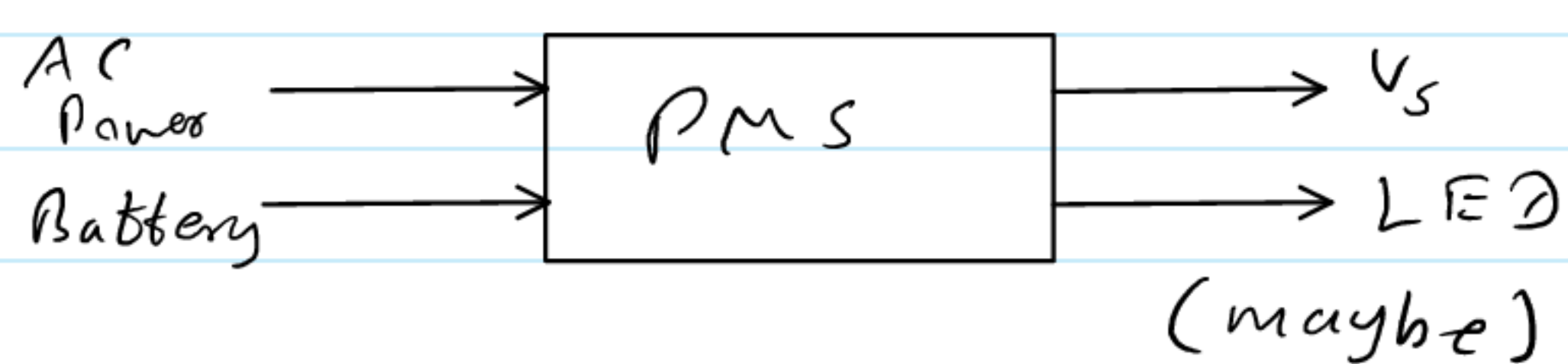
- Extracts the parameter of interest from the pulse coming from the photodiode (either peak or the integral)
- Must reset/update the output at each clock cycle

5. Comparison



- Must be able to handle hysteresis

6. PMS



- Must be capable of handling dual power modes
AC and backup battery
- LED (optional) may indicate anomalies